

TacoDepth: Towards Efficient Radar-Camera Depth Estimation with One-stage Fusion

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<https://github.com/RaymondWang987/TacoDepth>

Abstract

Radar-Camera depth estimation aims to predict dense and accurate metric depth by fusing input images and Radar data. Model efficiency is crucial for this task in pursuit of real-time processing on autonomous vehicles and robotic platforms. However, due to the sparsity of Radar returns, the prevailing methods adopt multi-stage frameworks with intermediate quasi-dense depth, which are time-consuming and not robust. To address these challenges, we propose TacoDepth, an efficient and accurate Radar-Camera depth estimation model with one-stage fusion. Specifically, the graph-based Radar structure extractor and the pyramid-based Radar fusion module are designed to capture and integrate the graph structures of Radar point clouds, delivering superior model efficiency and robustness without relying on the intermediate depth results. Moreover, TacoDepth can be flexible for different inference modes, providing a better balance of speed and accuracy. Extensive experiments are conducted to demonstrate the efficacy of our method. Compared with the previous state-of-the-art approach, TacoDepth improves depth accuracy and processing speed by 12.8% and 91.8%. Our work provides a new perspective on efficient Radar-Camera depth estimation.

1. Introduction

Radar-Camera depth estimation [6, 12, 20, 21, 25, 26, 28–31, 37] is a vital task for 3D perception and motion planning in autonomous driving [4, 15, 22, 49, 53] and robotics [13, 21, 39]. The task aims to predict dense metric depth by integrating images with mmWave Radar data, perceiving metric scale in different environments. As a typical range sensor on driving and robotic platforms, Radar is widely equipped and used for its high affordability and low power consump-

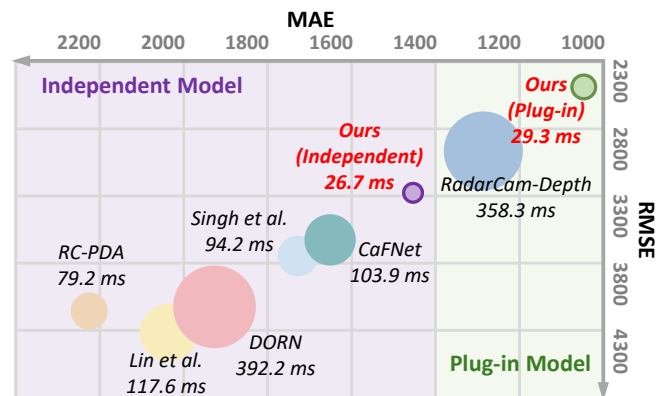


Figure 1. **Performance and efficiency.** Circle area indicates inference time (ms). Smaller circles showcase faster speed. The X-axis and Y-axis represent MAE and RMSE metrics for depth errors on the nuScenes [4] dataset. Lower MAE and RMSE mean higher accuracy. Our TacoDepth outperforms prior arts by large margins.

tion. It is valuable to explore Radar-Camera fusion in various tasks [2, 5, 9, 16, 32, 37] for its superior all-weather reliability. Besides, for real-time processing on vehicles and robotics, improving efficiency is crucial for fusion models.

The obstacle for fusing Radar in depth estimation lies in its sparsity and noises. Due to the large beam width, Radar point clouds are 1000x sparser with higher noise levels than LiDAR returns. To deal with the issue, the prevailing methods [21, 26, 28, 29, 31, 37, 38, 40] adopt complex multi-stage frameworks. Intermediate quasi-dense depth is predicted to bridge the gap between sparse Radar and final dense depth. For instance, two-stage methods [26, 28, 29, 31, 37, 38, 40] predict final depth based on intermediate results, as independent models without utilizing other depth predictors [3, 34, 35, 51]. RadarCam-Depth [21], as a plug-in module, converts relative depth [34, 35] to metric scale in four stages by post-processing. These multi-stage methods are inefficient and time-consuming, as presented in Fig. 1.

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Meanwhile, despite being described as quasi-dense, the intermediate depth [21, 26, 28, 29, 31, 37, 40] remains sparse and noisy. As shown in Fig. 2, only few pixels exhibit valid depth. For some adverse lighting or weather conditions, *e.g.*, glaring and nighttime scenarios, even no pixels are predicted with depth values. These defective intermediate results lead to disrupted structures, blurred details, or noticeable artifacts in final predictions, as shown in Fig. 4. With these observations, a question naturally arises: can models directly predict accurate and dense depth without the intermediate results, achieving one-stage Radar-Camera depth estimation with higher efficiency and robustness?

To answer the above question, we propose TacoDepth, an efficient and accurate Radar-Camera depth estimation framework with one-stage fusion. To simplify multi-stage fusion [21, 26, 28, 29, 31, 37, 38, 40] into one stage, TacoDepth needs to extract and integrate Radar information more effectively. Prior arts simply extract features from Radar coordinates [21, 37] or projected depth [26, 31], sensitive to noises and outliers [11]. In contrast, we utilize a graph-based Radar structure extractor to capture geometric structures and topologies of point clouds, improving robustness to inaccurate points [11, 18, 36, 43]. After that, our pyramid-based Radar fusion module integrates image and Radar features hierarchically. Shallow layers capture image details and Radar coordinates, while deep layers fuse scene semantics and structures [10, 14, 19, 41, 45, 50]. In each layer, Radar-centered flash attention mechanism is applied to establish the cross-modal correlations. For one Radar point, lightweight and accelerated flash attention [8] is calculated within Radar-centered areas for efficiency. Finally, TacoDepth predicts accurate and dense depth by a decoder.

Moreover, we further analyze the characteristics of different inference strategies in prior arts [21, 26, 28, 29, 31, 37, 38, 40]. Independent models [26, 28, 29, 31, 37, 40] are more efficient without initial depth prediction and post-processing, while the plug-in model [21] can be more accurate by leveraging cutting-edge relative depth predictors [3, 23, 34, 51]. Therefore, TacoDepth aims to be flexible for the two settings. An auxiliary and optional branch is added for initial depth, allowing TacoDepth to support both independent and plug-in processing. Our approach can better balance depth accuracy and model efficiency.

We conduct evaluations on nuScenes [4] and ZJU-4DRadarCam [21] datasets with different depth ranges, following prior arts [21, 31, 37]. As shown in Fig. 1, whether compared with independent [26, 28, 31, 37, 40] or plug-in [21] models, TacoDepth significantly outperforms prior arts [21, 26, 28, 31, 37, 40] in accuracy and efficiency. Besides, our approach achieves real-time processing of over 37 fps as an independent model and strong compatibility with various depth predictors [3, 34, 35, 51] as a plug-in model, further proving the efficiency and flexibility of TacoDepth.



Figure 2. **Intermediate quasi-dense depth maps [21, 26, 28, 29, 31, 37, 40] remain sparse and noisy.** Pixels with valid depth values are visualized by the gray areas in the red rectangular boxes.

Our main contributions can be summarized as follows:

- We propose TacoDepth, an efficient and accurate Radar-Camera depth estimation method with one-stage fusion.
- We design the pyramid-based Radar fusion module, fusing Radar and image features effectively and efficiently.
- TacoDepth is flexible for both independent and plug-in processing, balancing model efficiency and performance.

2. Related Work

MmWave Radar for Vehicles and Robotics. Autonomous driving [4, 15, 48, 53] and robotics [13, 21, 24, 39] rely on multiple sensors to perceive the environment, *e.g.*, camera, LiDAR, and mmWave Radar. LiDAR and Radar provide the metric scale of different scenarios as a complement to cameras. Compared with LiDAR, Radar is cheaper and more widely-equipped with lower power consumption. The reflection properties of mmWave also provide Radar with farther sensing ranges and stronger robustness to all-weather conditions. The main drawback of Radar lies in its sparsity and noise. Due to the large beam width, Radar point clouds are 1000x sparser [37] than LiDAR. Besides, Radar captures coordinates and depth values with lower accuracy. For instance, 3D Radar suffers from height ambiguity [31] due to insufficient antenna elements along the elevation axis, while this problem can be alleviated by the emerging 4D Radar. Thus, integrating Radar data is valuable for various tasks [2, 5, 9, 16, 27, 32, 37] in driving and robotic systems.

Radar-Camera Depth Estimation. The task aims to predict dense metric depth by merging images and Radar, facilitating 3D perception and motion planning of vehicles and robotics. Model efficiency is crucial for these real-time applications. However, to overcome the sparsity and noise of Radar, prior arts [21, 26, 28, 29, 31, 37, 38, 40] adopt inefficient multi-stage frameworks. The final depth is predicted based on intermediate quasi-dense depth. These methods can be divided into independent and plug-in models. Most independent models [26, 28, 29, 31, 37, 40] predict final depth in two stages without relying on other depth predictors [17, 33–35, 46, 47]. For example, Lin *et al.* [26] and DORN [28] use two CNNs [14] for quasi-dense and final depth. RC-PDA [31] and RC-DPT [29] convert intermediate results into multiple channels for varied confidence levels. Singh *et al.* [37] propose gated fusion to filter the inac-

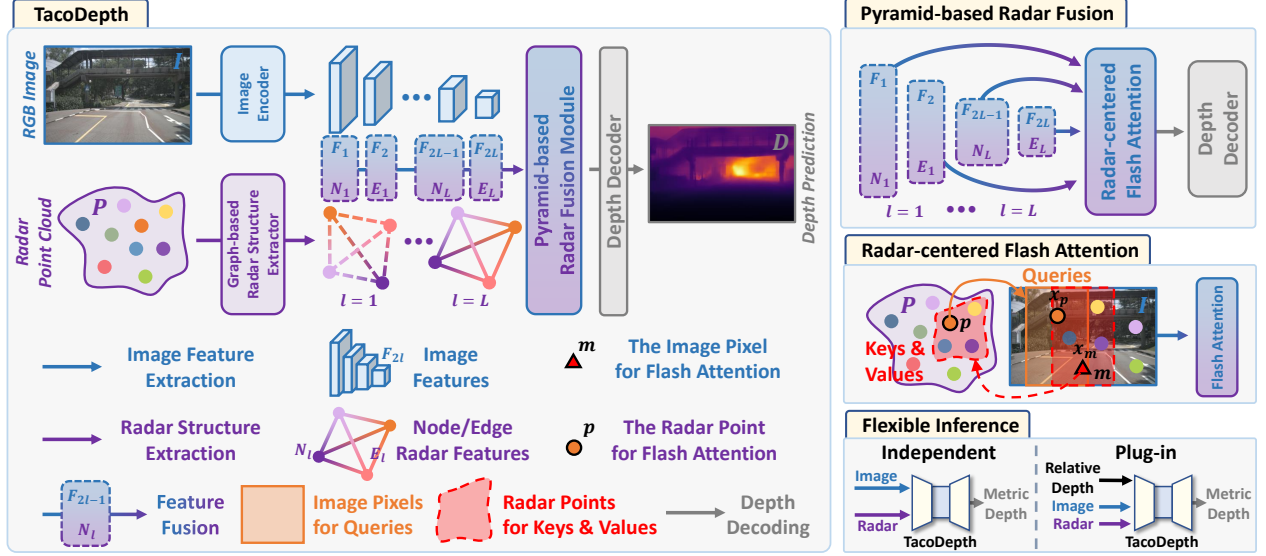


Figure 3. **Overview of the TacoDepth.** The graph-based Radar structure extractor captures graph structures of Radar point clouds through the node feature N_l and edge feature E_l in a certain layer l . The pyramid-based Radar fusion module integrates image and Radar features in a pyramidal hierarchical manner. In each layer, to efficiently build cross-modal correspondences, the Radar-centered flash attention is calculated within Radar-centered areas based on horizontal coordinates, e.g., pixels in the orange area as queries, while Radar points in the red area as keys and values. Our TacoDepth achieves efficient and accurate Radar-Camera depth estimation in one stage. During inference, the model is flexible and supports both independent and plug-in processing, facilitating a better balance of model efficiency and accuracy.

curate quasi-dense depth. Besides, RadarCam-Depth [21] designs a plug-in module, aligning relative depth [34, 35] to metric scale in four stages, including initial depth, global scale alignment, quasi-dense depth, and scale learner. These methods [21, 26, 28, 29, 31, 37, 38, 40] are inefficient with complex stages. Defective intermediate depth also limits their robustness. In contrast, we propose TacoDepth model to achieve efficient, accurate, and flexible one-stage fusion.

3. TacoDepth

We present a detailed illustration of TacoDepth, an efficient Radar-Camera depth estimation framework with one-stage fusion. In Sec. 3.1, we provide an overall description to outline our approach. In Sec. 3.2, we elaborate on the detailed architecture of the pyramid-based Radar fusion module. TacoDepth can be flexible for independent and plug-in processing, which will be discussed in Sec. 3.3. Besides, the training procedure for TacoDepth is illustrated in Sec. 3.4.

3.1. Overview

Here we present an overview of our TacoDepth framework. Given a single image $I \in \mathcal{R}^{H \times W \times 3}$ and a Radar point cloud $P \in \mathcal{R}^{K \times 3}$ as inputs, our goal is to predict dense metric depth $D \in \mathcal{R}^{H \times W}$. H and W represent image height and weight. The point cloud P contains K points $p \in \mathcal{R}^3$. Radar-Camera depth estimation can be divided into two primary subtasks: the extraction and the integration of Radar information. Accordingly, as depicted in Fig. 3,

TacoDepth comprises a graph-based Radar structure extractor and a pyramid-based Radar fusion module to undertake these two important steps respectively.

Firstly, for Radar feature extraction, previous methods [21, 37] simply extract features from the coordinates of individual Radar points, focusing solely on the point-level information. For example, Singh *et al.* [37] extract point features in 32,256 dimensions from the three-dimensional coordinates using a multi-layer perceptron (MLP), which inevitably introduces substantial redundancy and noise.

To achieve efficient Radar-Camera fusion in a single stage, TacoDepth needs to extract more comprehensive and effective Radar information, prompting us to reconsider the complementarity between camera images and Radar data. Compared to the coordinates, the geometric structures of point clouds, e.g., the distances or relations between Radar points, are potentially more informative for Radar-Camera fusion. The overall topologies are also more robust to inaccurate outliers [11]. To this end, we employ the graph-based Radar structure extractor with a lightweight GNN [11, 36, 43] architecture. Specifically, Radar points are regarded as the graph nodes. The adjacency matrix of all points represents the graph edges. In each layer, the node features of Radar points can be updated and aggregated along edges by PCA-GM [44]. The graph-based Radar structure extractor can capture point coordinates and graph structures from shallow layers to deep layers [11, 41, 45].

After extracting the Radar structures, the subsequent step

is to merge RGB and Radar features. Prior arts [21, 26, 28, 29, 31, 37, 38, 40] adopt multi-stage methods with intermediate quasi-dense depth, which are inefficient and not robust as illustrated in Sec. 1. In contrast, we propose our pyramid-based Radar fusion module for efficient one-stage fusion.

To be specific, features of the two modalities are integrated in a pyramidal hierarchical manner. Shallow layers can merge image details with Radar coordinates, whereas deeper layers could fuse scene semantics and geometric structures [10, 14, 19, 41, 45, 50]. In each layer, we leverage Radar-centered flash attention for cross-modal correlations and feature fusion. The queries are derived from image pixels, while keys and values originate from Radar points. For each point, we delineate Radar-centered areas based on the horizontal coordinates, as exemplified by the orange and red areas in Fig. 3, given that the projected horizontal coordinates of Radar points are generally more precise than the height dimension [31, 37]. Lightweight flash attention [8] is calculated within these areas to maintain efficiency, since external pixels and points can not be correlated. Finally, with the fused features from all layers, TacoDepth predicts accurate and dense metric depth by a depth decoder [37].

Furthermore, TacoDepth is adaptable for both independent and plug-in inference. An optional input branch is introduced to handle the initial relative depth. For the independent inference, our model achieves real-time processing of over 37 fps. For the plug-in manner, TacoDepth demonstrates compatibility with various cutting-edge depth predictors [3, 34, 35, 51], as proved in Sec. 4.5. Consequently, our approach can deliver a balanced trade-off between performance and efficiency for different users and applications.

3.2. Pyramid-based Radar Fusion

For the Radar point cloud P , our graph-based Radar structure extractor derives node feature N_l and edge feature E_l at a certain layer $l \in \{1, 2, \dots, L\}$, where L represents the layer number of the GNN module [11, 36, 43]. For the input image I , the image encoder [14, 21, 37] extracts multi-scale features $\{F_1, F_2, \dots, F_{2L}\}$. The pyramid-based Radar fusion module hierarchically integrates the multi-modal features. N_l and E_l will be fused with image features in adjacent layers. We merge the node feature N_l with image feature F_{2l-1} , and the edge feature E_l with F_{2l} , respectively.

The prerequisite for feature fusion is cross-modal correspondences between Radar points and RGB pixels. Due to the noisy elevation coordinates of 3D Radar, prior arts [21, 29, 31, 37, 40] try to build one-to-many mappings by the intermediate quasi-dense depth. They adopt explicit classification supervision with the depth discrepancies between LiDAR and Radar as pseudo-labels. However, the quasi-dense depth prediction leads to inefficiency and erroneous final outcomes, as discussed in Sec. 1. Given the absence of reliable explicit labels and supervision, we hope our TacoDepth

can learn and establish the correspondences implicitly.

Therefore, for efficient Radar-Camera correlations, we propose the Radar-centered flash attention in TacoDepth. We define Radar-centered areas based on the precise horizontal coordinates of Radar points and image pixels. For each point, the attention is computed only with image pixels in the area, since external pixels can not be correlated. Here we take $E_l \in \mathcal{R}^{K \times K}$ and $F_{2l} \in \mathcal{R}^{h \times w \times C_l}$ as examples, where h and w are feature height and width. K and C_l refer to the point number and layer feature channel. For a Radar point p with the horizontal coordinate x_p , as showcased by the exemplary orange area in Fig. 3, we only retain image pixels m whose coordinates x_m are close to x_p :

$$\hat{F}_{2l} = F_{2l} [x_p - a_l < x_m < x_p + a_l], \hat{F}_{2l} \in \mathcal{R}^{\hat{M} \times C_l}, \quad (1)$$

where a_l indicates the width of the Radar-centered area in layer l . The notation $[\cdot]$ depicts the fetching operation of pixels or points. $\hat{F}_{2l}$ is the image feature of the remaining $\hat{M}$ pixels to generate queries. For a random pixel m in $\hat{F}_{2l}$, as represented by the red area in Fig. 3, the attention calculation is also restricted to adjacent Radar points:

$$\hat{E}_l = E_l [x_m - a_l < x_p < x_m + a_l], \hat{E}_l \in \mathcal{R}^{\hat{K} \times K}. \quad (2)$$

The feature dimension maintains the original K to preserve integral graph topologies. $\hat{E}_l$ is the edge feature of the retained $\hat{K}$ points for keys and values. In this way, the computational costs can be significantly decreased with the small subset of $\hat{M}$ pixels and $\hat{K}$ points. For the pixel m , we generate queries from $\hat{F}_{2l}[m] \in \mathcal{R}^{C_l}$, while keys and values are derived from $\hat{E}_l$. Radar-centered attention is computed as:

$$F'_{2l}[m] = \text{softmax} \frac{W_q \hat{F}_{2l}[m] (W_k \hat{E}_l)^T}{\sqrt{C_l}} W_v \hat{E}_l, \quad (3)$$

in which W_q , W_k , and W_v are learnable linear projections. The fused Radar-Camera feature $F'_{2l} \in \mathcal{R}^{h \times w \times C_l}$ can be acquired for the current layer. The above calculation is incorporated into attention blocks with MLPs and residual connections. To further improve model efficiency, we employ lightweight and accelerated flash attention modules [7, 8] to replace the original attention computation [42].

Similarly, the node feature N_l can be merged with the image feature F_{2l-1} by slightly adjusting the dimensions of linear projections. Ultimately, the acquired Radar-Camera features $\{F'_1, F'_2, \dots, F'_{2L}\}$ are fed to a common depth decoder [21, 37] to predict accurate metric depth.

As a consequence, the proposed pyramid-based Radar fusion module enables efficient and effective Radar-Camera fusion in a single stage. The Radar-centered flash attention establishes cross-modal correlations with minimal computational costs. Without relying on the previous multi-stage frameworks and intermediate quasi-dense depth [21, 26, 28, 29, 31, 37, 38, 40], TacoDepth achieves higher efficiency and stronger robustness for Radar-Camera depth estimation.

3.3. Flexible Inference

Considering the varied inference manners in prior arts [21, 26, 28, 29, 31, 37, 38, 40], the plug-in module [21] could achieve higher accuracy using state-of-the-art relative depth predictors [34, 35, 52], while independent models [26, 28, 29, 31, 37, 38, 40] are capable of attaining faster inference speed without initial depth prediction and post-processing.

Therefore, TacoDepth is designed to be flexible, supporting both independent and plug-in predictions. For the plug-in inference, an initial relative depth map is converted to metric scale by fusing camera and Radar information. If we denote the TacoDepth model as $\mathcal{T}_\theta$, parameterized by θ , the depth prediction process can then be formulated as follows:

$$D = \mathcal{T}_\theta(I, P \mid D^*), \quad (4)$$

where the vertical bar ($\mid$) separates the required and optional inputs. D is the predicted depth of our model. D^* is the optional input of initial relative depth from depth predictors [3, 34, 35, 51]. We introduce an auxiliary branch to process and fuse D^* with image features, while keeping all other settings identical to the independent mode. Our plug-in manner demonstrates compatibility with various depth predictors, *e.g.*, MiDaS [3, 34], DPT [35], and Depth-Anything-v2 [51]. In this way, TacoDepth can achieve a better balance of model efficiency and accuracy, offering superior flexibility for different users and applications.

3.4. Training the TacoDepth

In the training phase, the training data [4, 21] is randomly divided into two equal portions in each epoch. For the auxiliary branch, one half of the data is provided with initial relative depth, while the other half adopts zero as input. This training strategy facilitates the simultaneous optimization of the independent inference and plug-in prediction.

As for the training loss, we employ the same depth ground truth and L_1 loss as the prior art [37]. Specifically, LiDAR-captured depth D_{gt} and reprojection-accumulated depth D_{acc} are utilized for supervision [21, 31, 37]. The overall loss function ℓ_{L_1} can be expressed as follows:

$$\ell_{L_1} = \frac{1}{|\Omega_{gt}|} \sum_{\Omega_{gt}} |D - D_{gt}| + \frac{\lambda}{|\Omega_{acc}|} \sum_{\Omega_{acc}} |D - D_{acc}|, \quad (5)$$

where Ω_{gt} and Ω_{acc} depict the masks of D_{gt} and D_{acc} with valid depth values. λ is the coefficient of the second term.

4. Experiments

To prove the efficacy of our TacoDepth, we conduct experiments on the nuScenes [4] and ZJU-4DRadarCam [21] datasets following prior arts [20, 21, 26, 28, 29, 31, 37, 40]. In Sec. 4.1, we describe the experimental datasets and evaluation protocols. The implementation details of our model

are illustrated in Sec. 4.2. In Sec. 4.3, we compare the TacoDepth with previous Radar-Camera depth estimation approaches to demonstrate our state-of-the-art performance. The inference speed and model efficiency of different methods are evaluated and discussed in Sec. 4.4. In Sec. 4.5, we perform ablation studies to expound on our specific designs.

4.1. Datasets and Evaluation Protocols

Due to the scarcity of public datasets for Radar-Camera depth estimation, prior works [20, 26, 28, 29, 31, 37, 38, 40] only conduct experiments on nuScenes dataset [4] with 3D Radar. Recently, RadarCam-Depth [21] releases their ZJU-4DRadarCam dataset [21] with 4D Radar to alleviate this problem. We adopt these two datasets [4, 21] for experiments, proving our effectiveness on both 3D and 4D Radar.

NuScenes Dataset. As an outdoor driving dataset, the nuScenes [4] is collected by vehicles equipped with cameras, 3D Radar, LiDAR, and IMU. It contains 1,000 scenes and around 40,000 synchronized Radar-Camera keyframes captured in Boston and Singapore. We follow the same train-test split as prior arts [21, 26, 29, 31, 37, 40], with 700 scenes for training, 150 for validation, and 150 for testing.

ZJU-4DRadarCam Dataset. The dataset [21] is collected by a ground robot fitted with more advanced camera, 4D Radar, and LiDAR sensors, offering denser Radar data and LiDAR depth than the nuScenes [4] dataset. It comprises 33,409 Radar-Camera keyframes in total. We follow the official data split [21] with 29,312 frames for training and validation, and 4,097 samples for testing in our experiments.

Evaluation Protocols. We evaluate independent and plug-in models separately for fair comparisons on the two datasets [4, 21] with depth ranges of 50, 70, and 80 meters [21, 31, 37, 40]. On nuScenes [4], previous state-of-the-art methods [12, 20, 21, 26, 28, 31, 37, 40] are compared comprehensively to the best of our knowledge. For the recently published ZJU-4DRadarCam [21], since most prior works do not include experiments on the dataset, we can only follow the same comparative methods [21, 28, 37] and results as RadarCam-Depth [21]. Besides, we also evaluate plug-in models with different depth predictors [3, 35, 51] to showcase model flexibility on the ZJU-4DRadarCam [21].

Evaluation Metrics. We adopt the commonly-applied depth metrics MAE, RMSE, iMAE, iRMSE, δ_1 , and *Rel* as prior arts [21, 26, 31, 37]. See supplement for more details.

4.2. Implementation Details

Data Processing. We adhere to the data processing procedures of prior arts [21, 37]. For the nuScenes [4], the D_{acc} in Eq. 5 is generated by accumulating the D_{gt} of 160 nearby frames [21, 37]. For the ZJU-4DRadarCam dataset [21], the D_{acc} is obtained via the linear interpolation [1, 21] of D_{gt} .

Model Architecture. In line with prior arts [21, 37], the ResNet-18 [14] is adopted as the image encoder. The graph-

Type	Method	Radar Frames	Images	Time(ms)↓	0 - 50m		0 - 70m		0 - 80m	
					MAE↓	RMSE↓	MAE↓	RMSE↓	MAE↓	RMSE↓
Independent	Lin <i>et al.</i> [26] (IROS'20)	3	1	117.6	2034.9	4316.5	2294.7	5338.2	2371.0	5623.0
	RC-PDA [31] (CVPR'21)	5	3	79.2	2225.0	4156.5	3326.1	6700.6	3713.6	7692.8
	R4Dyn [12] (3DV'21)	4	1	—	—	—	—	—	—	6434.0
	DORN [28] (ICIP'21)	5(×3)	1	392.5	1926.6	4124.8	2380.6	5252.7	2467.7	5554.3
	Singh <i>et al.</i> [37] (CVPR'23)	1	1	94.2	1727.7	3746.8	2073.2	4590.7	2179.3	4898.7
	CaFNet [40] (IROS'24)	1	1	103.9	1674.2	3674.5	2010.3	4493.1	2109.8	4765.6
	Li <i>et al.</i> [20] (ECCV'24)	1	1	<u>67.6</u>	<u>1524.5</u>	<u>3567.3</u>	<u>1822.9</u>	<u>4303.6</u>	<u>1927.0</u>	<u>4609.6</u>
	TacoDepth (Ours)	1	1	26.7	1423.6	3275.8	1712.6	3960.5	1833.4	4150.2
Plug-in	RadarCam-Depth [21] (ICRA'24)	1	1	<u>358.3</u>	<u>1286.1</u>	<u>2964.3</u>	<u>1587.9</u>	<u>3662.5</u>	<u>1689.7</u>	<u>3948.0</u>
	TacoDepth (Ours)	1	1	29.3	1046.8	2487.5	1347.1	3152.8	1492.4	3324.8

Table 1. **Comparisons with state-of-the-art methods on the nuScenes dataset [4] (in millimeters).** Independent [12, 20, 26, 28, 31, 37, 40] and plug-in [21] models are evaluated separately, with maximum evaluation distances of 50, 70, and 80 meters as prior arts [21, 37]. Plug-in models are compared using the same depth predictor DPT-Hybrid [35] as RadarCam-Depth [21]. We report the average runtime of processing one 900×1600 frame and 30 Radar points by different methods on one NVIDIA RTX A6000 GPU. The inference time of plug-in models does not contain initial depth prediction. Some methods [12, 26, 28, 31] require multiple frames and Radar scans for inference, while others [20, 21, 37, 40] only need a single frame as input. The best performance is in boldface. Second best is underlined.

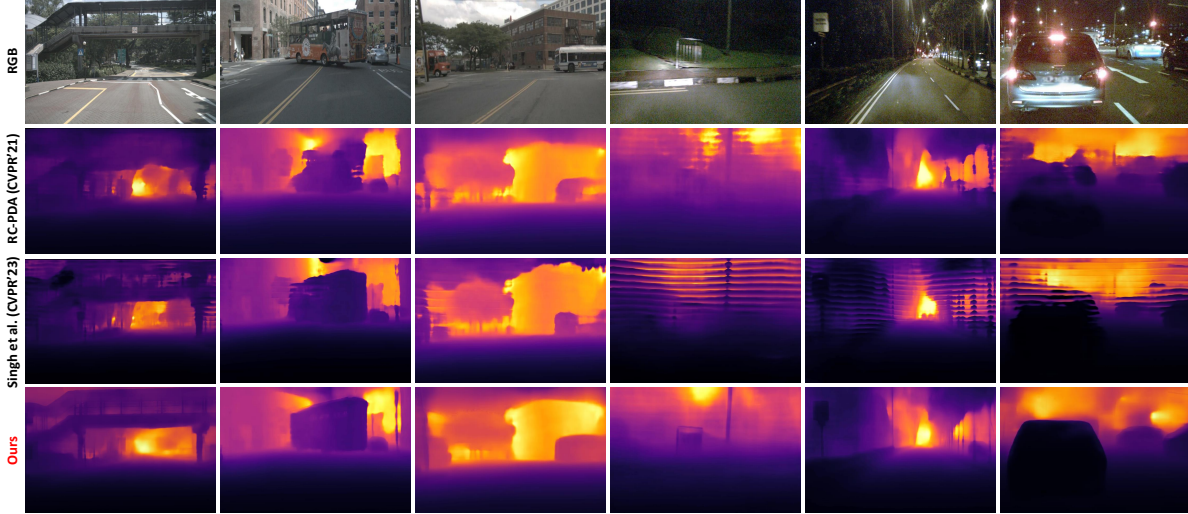


Figure 4. **Visual results of independent models on nuScenes [4].** Both daytime and nighttime samples are presented. Prior arts [31, 37] exhibit disrupted structures and noticeable artifacts. TacoDepth produces accurate depth with finer details and more complete structures.

based Radar structure extractor contains $L = 3$ layers with the GNN [11, 36] architecture. The widths a_l of Radar-centered areas in Eq. 1 and Eq. 2 are set to $\{48, 32, 16\}$ for the three layers. Refer to the supplement for more details.

Training Recipe. The model is trained by Adam optimizer on two NVIDIA A6000 GPUs with a batch size of 12 for 50 epochs. The initial learning rate is $1e^{-4}$ and decreases by $1e^{-5}$ for every ten epochs. The input image size is 900×1600 [21, 37] and 300×1280 [21] on nuScenes [4] and ZJU-4DRadarCam [21], respectively. The λ in Eq. 5 is set to one. All other settings are consistent with prior works [21, 37].

4.3. Comparisons with State-of-the-art Methods

Comparisons with Independent Models. Independent models [12, 20, 26, 28, 29, 31, 37, 38, 40] predict dense

metric depth without using relative depth predictors [3, 34, 35, 51]. Most of them adopt two-stage methods with intermediate quasi-dense depth, which are inefficient and not robust. Our TacoDepth significantly surpasses these methods in efficiency and accuracy with one-stage fusion. For example, in Table 1, compared with the recent CaFNet [40], our model reduces RMSE, MAE, and inference time by over 10.9%, 13.1%, and 74.3% on nuScenes dataset [4]. Additionally, Li *et al.* [20] aim to improve the training paradigm under sparse supervision by their Disruption-Compensation framework. Without using such advanced training and supervision strategy, TacoDepth still outperforms Li *et al.* [20] by over 60.5%, 4.9%, and 8.1% for runtime, MAE, and RMSE, showing the efficacy of our proposed architectures.

Similarly, in Table 2, compared with Singh *et al.* [37],

Type	Method	0 - 50m				0 - 70m				0 - 80m			
		MAE↓	RMSE↓	iMAE↓	iRMSE↓	MAE↓	RMSE↓	iMAE↓	iRMSE↓	MAE↓	RMSE↓	iMAE↓	iRMSE↓
Independent	DORN [28] (ICIP'21)	2210.2	4129.7	19.8	<u>31.9</u>	2402.2	4625.2	19.8	<u>31.9</u>	2447.6	4760.0	19.9	<u>31.9</u>
	Singh <i>et al.</i> [37] (CVPR'23)	<u>1785.4</u>	3704.6	<u>18.1</u>	35.3	<u>1932.7</u>	<u>4137.1</u>	18.0	35.2	<u>1979.5</u>	4309.3	17.9	35.1
	TacoDepth (Ours)	1120.1	2686.7	12.8	25.0	1181.8	2906.3	12.7	24.9	1201.1	2990.7	12.7	24.9
Plug-in	RadarCam-Depth [21] (ICRA'24)	<u>1067.5</u>	<u>2817.4</u>	<u>10.5</u>	<u>22.9</u>	<u>1157.0</u>	<u>3117.7</u>	<u>10.4</u>	<u>22.9</u>	<u>1183.5</u>	<u>3229.0</u>	<u>10.4</u>	<u>22.8</u>
	TacoDepth (Ours)	930.2	2477.3	9.3	20.8	983.1	2779.6	9.3	20.9	1032.5	2850.3	9.4	20.9

Table 2. **Comparisons with state-of-the-art methods on ZJU-4DRadarCam [21] (in millimeters).** Plug-in models are compared using the same relative depth predictor DPT-Hybrid [35]. Since many prior works do not include experiments on this new dataset, we adhere to the same compared methods [21, 28, 37] and metrics as RadarCam-Depth [21]. Best performance is in boldface. Second best is underlined.



Figure 5. **Visual results of plug-in models [21] on the ZJU-4DRadarCam [21].** The same depth predictor DPT-Hybrid [35] is adopted for fair comparisons. Regions with obvious differences are highlighted in the rectangular boxes. Best view zoomed in on-screen for details.

TacoDepth reduces depth errors by over 37.3%, 27.5%, 28.7%, and 29.1% for MAE, RMSE, iMAE, and iRMSE on ZJU-4DRadarCam [21]. The superior efficiency and accuracy prove the effectiveness of our one-stage model design.

Comparisons with Plug-in Models. Plug-in models [21] convert relative depth to metric depth by fusing the RGB and Radar data. Our method supports both independent and plug-in processing by the auxiliary input branch. For the plug-in mode, in Table 1, TacoDepth decreases the MAE and RMSE by over 11.7% and 13.9% than RadarCam-Depth [21] on nuScenes [4]. Owing to their complex four-stage framework [21], we even improve the speed by 91.8%. Similarly, in Table 2, TacoDepth reduces the MAE, RMSE, iMAE, and iRMSE by over 12.8%, 10.8%, 9.6%, and 8.3% on ZJU-4DRadarCam [21]. The results prove the generality and flexibility of TacoDepth, providing a better balance of model efficiency and accuracy for different users.

To be mentioned, in the plug-in mode, TacoDepth can

be compatible with various relative depth predictors [3, 35, 51]. It can benefit from stronger depth models [3, 51] to achieve higher accuracy, which will be presented in Table 4.

Qualitative Comparisons. The visual results of independent models [31, 37] are showcased in Fig. 4. Both daytime and nighttime samples are compared. TacoDepth can predict accurate depth maps with more complete structures (*e.g.*, the cars and buses) and more meticulous details (*e.g.*, the pedestrian overpass and metal guardrail). The plug-in models [21] are compared in Fig. 5. Based on the same depth predictor DPT-Hybrid [35], our method produces noticeably better structures and details in the rectangular boxes. The improvements are attributed to our graph-based Radar structure extractor and pyramid-based Radar fusion module, which extract and fuse Radar structures effectively.

Model Robustness. Multi-stage methods [26, 28, 31, 37, 40] predict quasi-dense depth as an intermediate step. However, as shown in Fig. 2, the intermediate results are still

Type	Method	Params (M) $\downarrow$	FLOPs (G) $\downarrow$	Time (ms) $\downarrow$
Indep- endent	DORN [28] (ICIP'21)	183.38	923.47	392.5
	Singh <i>et al.</i> [37] (CVPR'23)	22.81	502.09	94.2
	TacoDepth (Ours)	13.47	139.30	26.7
Plug-in	RadarCam-Depth [21] (ICRA'24)	33.26	619.02	358.3
	TacoDepth (Ours)	14.25	139.87	29.3

Table 3. **Model efficiency.** We report FLOPs, model parameters, and runtime of different methods. Since varied depth predictors can be applied, the results of plug-in models do not contain initial depth prediction. FLOPs and runtime are evaluated with one 900 $\times$ 1600 image and 30 Radar points on one NVIDIA A6000 GPU.

	Initial			RadarCam-Depth [21]			TacoDepth (Ours)		
	$\delta_1 \uparrow$	Rel $\downarrow$	MAE $\downarrow$	$\delta_1 \uparrow$	Rel $\downarrow$	MAE $\downarrow$	$\delta_1 \uparrow$	Rel $\downarrow$	MAE $\downarrow$
DPT-Hybrid [35]	0.903	0.092	—	0.921	0.087	1157.0	0.932	0.076	983.1
MiDaS-v3.1 [3]	0.917	0.081	—	0.926	0.086	1177.2	0.944	0.071	852.4
Depth-Anything-v2 [51]	0.932	0.069	—	0.940	0.073	924.7	0.961	0.059	730.2

Table 4. **Plug-in models with varied depth predictors.** Our method outperforms RadarCam-Depth [21] with different depth predictors [3, 35, 51] on ZJU-4DRadarCam [21] (0-70 meters).

sparse and noisy, especially on challenging nighttime and glaring scenes, leading to disrupted structures, blurred details, and obvious artifacts in their final depth predictions, as presented in Fig. 4. In contrast, we propose the one-stage TacoDepth without the quasi-dense depth. By effectively integrating the complementary image details and Radar structures, our model produces robust depth for both daytime and nighttime scenarios in Fig. 4. See supplement for more quantitative and visual results on nighttime scenes.

4.4. Model Efficiency

We evaluate the processing time of different methods in Table 1. Compared with the second-best independent model of Li *et al.* [20] and the plug-in model RadarCam-Depth [21], TacoDepth improves inference speed by 60.5% and 91.8%.

To further showcase our efficiency, we report model parameters and FLOPs in Table 3. Compared with the two-stage Singh *et al.* [37], the one-stage TacoDepth reduces the parameters and FLOPs by 40.9% and 72.2%. Compared with the four-stage plug-in model RadarCam-Depth [21], we decrease parameters and FLOPs by 57.1% and 77.4%.

Note that, the differences between the plug-in and independent modes of our method arise from the auxiliary branch to fuse initial relative depth. TacoDepth achieves real-time processing of 37.5 fps by independent inference.

4.5. Ablation Studies

Plug-in Inference. In Table 4, our TacoDepth outperforms the RadarCam-Depth [21] with three different relative depth predictors [3, 35, 51]. Using more advanced depth predictors, *e.g.*, MiDaS-v3.1 [3] and Depth-Anything-v2 [51], can further boost our accuracy without extra effort. The results demonstrate the versatility and flexibility of our approach.

Efficacy of TacoDepth. In Table 5, we ablate the graph-

Method	RGB	Radar	GE	PF	MAE $\downarrow$	RMSE $\downarrow$
RGB Baseline	✓	✗	✗	✗	2474.3	5402.1
Singh <i>et al.</i> [37] (CVPR'23)	✓	✓	✗	✗	2073.2	4590.7
TacoDepth (<i>w/o</i> GE)	✓	✓	✗	✓	<u>1815.6</u>	<u>4189.8</u>
TacoDepth (Ours)	✓	✓	✓	✓	1712.6	3960.5

Table 5. **Efficacy of the TacoDepth.** We ablate the effectiveness of graph-based Radar structure extractor (GE) and pyramid-based Radar fusion (PF) for Radar structure extraction and integration. We adopt the same baseline as Singh *et al.* [37]. TacoDepth (*w/o* GE) uses the same MLP to extract Radar point features as prior art [37]. The results are evaluated on nuScenes [4] in 0-70 meters.

Metric	$L = 1$	$L = 2$	$L = 3$	$L = 4$
MAE $\downarrow$	1840.6	1787.8	1712.6	1729.3
RMSE $\downarrow$	4203.7	4046.9	3960.5	3984.2

Table 6. **Layer number L .** Results are on nuScenes [4] in 0-70m.

based Radar structure extractor (GE) and pyramid-based Radar fusion module (PF). With the same MLP to extract Radar point features [37], TacoDepth (*w/o* GE) reduces MAE and RMSE by 12.4% and 8.7% than Singh *et al.* [37], which are attributed to the PF. Replacing the MLP [37] with GE further decreases MAE and RMSE by 5.7% and 5.5%.

We adopt the RGB baseline of Singh *et al.* [37], a common single-image depth U-Net. Overall, TacoDepth improves accuracy beyond the baseline by 30.8% and 26.7% (MAE and RMSE), proving the efficacy of our approach.

Layer Number L . In Table 6, we ablate the layer number L of GE and PF. Compared with shallow layers ($L = 1, 2$), deeper layers ($L = 3$) better capture structures of point clouds [41, 45], producing higher accuracy. Due to Radar sparsity, more layers ($L = 4$) does not yield improvements but decreases efficiency. We adopt $L = 3$ in experiments.

5. Conclusion

In this paper, we propose TacoDepth, an efficient and accurate Radar-Camera depth estimation model with one-stage fusion. Different from the multi-stage methods with quasi-dense depth, TacoDepth utilizes the graph-based Radar structure extractor and pyramid-based Radar fusion module to capture and integrate Radar features, achieving superior efficiency and robustness. Moreover, TacoDepth is flexible for independent and plug-in processing, offering a better balance of speed and accuracy. Our work provides a new perspective on the efficient Radar-Camera depth estimation.

Limitations. We currently provide one efficient implementation of our TacoDepth. In future work, more techniques can be explored for varied performances and applications.

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